ArraySim5-4: Combined Site Heterogeneity
Data-Generating Process, Oracle Truth, and CCW Comparisons

1 Purpose

ArraySim5-4 evaluates federated clone–censor–weight (Fed-CCW) estimation under three sources of between-site heterogeneity occurring simultaneously:

1. unequal site sample sizes;
2. different site-specific distributions of measured baseline covariates; and
3. different residual site treatment-initiation practices after conditioning on measured patient history.

Each low, moderate, or high scenario specifies all three components jointly. The total sample size remains 5,000, but its distribution across five sites becomes increasingly unequal. Patient-mix separation and residual initiation practice separation increase at the same time.

The simulation compares Fed-CCW with two pooled CCW implementations. The first pooled implementation uses fully site-stratified nuisance models and is mathematically aligned with Fed-CCW. The second uses a conventional pooled logistic initiation model with site fixed effects and one common covariate slope vector, thereby borrowing information across sites.

2 Indices, observed data, and simulation grid

Sites are indexed by $k \in \{1, \ldots, 5\}$, patients by i, and discrete follow-up intervals by $m = 1, \ldots, M$, where $M = 25$. For patient i at site k, let

- X_{1ik} and X_{2ik} denote baseline covariates;
- L_{1ikm} and L_{2ikm} denote time-varying covariates;
- S_{ik} denote the treatment-initiation interval;
- T_{ik} denote the terminal-event interval; and
- $A^{\tau}_{ik} = \mathbb{I}(S_{ik} \le \tau)$ indicate initiation within the grace period.

If initiation or the event does not occur during follow-up, its time is coded as $M + 1$.

The simulation grid is

$$\tau \in \{5, 10, 15\}, \qquad \beta_{\text{trt}} \in \{-1.0, -0.7, -0.5\}, \qquad h \in \{\text{low}, \text{moderate}, \text{high}\},$$

giving $3 \times 3 \times 3 = 27$ array tasks. Each task contains 100 Monte Carlo replicates.

3 Joint site-heterogeneity settings

For heterogeneity level h, define

$$\mathcal{G}_h = \{n_{kh}, \mu_{kh}, \sigma_{kh}, p_{kh}, \alpha_{kh} : k = 1, \ldots, 5\},$$

where n_{kh} is site size, $X_1 \sim N(\mu_{kh}, \sigma_{kh}^2)$, $X_2 \sim \text{Bernoulli}(p_{kh})$, and α_{kh} is the site-specific treatment-initiation intercept.

Table 1: Combined site-heterogeneity settings.

Level	Site	n_{kh}	X_1 distribution	X_2 probability	α_{kh}
Low	1	900	$N(-0.250, 0.950^2)$	0.350	-3.250
	2	950	$N(-0.125, 0.975^2)$	0.375	-3.125
	3	1000	$N(0, 1^2)$	0.400	-3.000
	4	1050	$N(0.125, 1.025^2)$	0.425	-2.875
	5	1100	$N(0.250, 1.050^2)$	0.450	-2.750
Moderate	1	600	$N(-0.750, 0.850^2)$	0.200	-3.750
	2	800	$N(-0.375, 0.925^2)$	0.300	-3.375
	3	1000	$N(0, 1^2)$	0.400	-3.000
	4	1200	$N(0.375, 1.075^2)$	0.500	-2.625
	5	1400	$N(0.750, 1.150^2)$	0.600	-2.250
High	1	400	$N(-1.500, 0.700^2)$	0.100	-4.500
	2	600	$N(-0.750, 0.850^2)$	0.250	-3.750
	3	800	$N(0, 1^2)$	0.400	-3.000
	4	1200	$N(0.750, 1.150^2)$	0.550	-2.250
	5	2000	$N(1.500, 1.300^2)$	0.700	-1.500

For every level,

$$\sum_{k=1}^{5} n_{kh} = 5000.$$

The maximum-to-minimum site-size ratios are approximately 1.22, 2.33, and 5.00 for the low, moderate, and high levels. The X_1 mean ranges are 0.5, 1.5, and 3.0; the X_2 probability ranges are 0.1, 0.4, and 0.6; and the initiation-intercept ranges are 0.5, 1.5, and 3.0.

The gradients are intentionally aligned. Larger sites tend to enroll patients with larger X values and have higher residual initiation propensity. Thus, high heterogeneity is a joint stress test rather than three independent one-factor perturbations.

4 Data-generating process

4.1 Baseline covariates

At level h, site k generates

$$X_{1ik} \sim N(\mu_{kh}, \sigma_{kh}^2), \qquad X_{2ik} \sim \text{Bernoulli}(p_{kh}).$$

Because n_{kh} also varies, the marginal target-population patient mix is the size-weighted five-site mixture rather than an equally weighted mixture.

4.2 Time-varying covariates

For $\ell \in \{1, 2\}$,

$$L_{\ell,ikm} = 0.6L_{\ell,ik,m-1} + 0.3X_{1ik} + 0.2X_{2ik} + \varepsilon_{\ell,ikm}, \qquad \varepsilon_{\ell,ikm} \sim N(0, 1), \tag{1}$$

with $L_{\ell,ik0} = 0$. Treatment does not affect subsequent L. Covariates are generated only while the patient remains alive.

4.3 Treatment initiation

Among patients alive and untreated at the beginning of interval m, define

$$q_{ikm} = \Pr(S_{ik} = m \mid S_{ik} \geq m, T_{ik} \geq m, X_{ik}, L_{ikm}, K_i = k).$$

The initiation DGP is

$$\operatorname{logit}(q_{ikm}) = \alpha_{kh} + 0.8X_{1ik} - 0.3X_{2ik} + 0.5L_{1ikm} + 0.4L_{2ikm}. \tag{2}$$

The covariate slopes are common across sites and heterogeneity levels. Only the site intercept α_{kh} changes. Consequently, two patients with the same measured history can have different initiation probabilities because they receive care at different sites.

4.4 Terminal-event process

Conditional on being alive at interval m, the event hazard is

$$\begin{aligned}\operatorname{logit}\Pr(T_{ik} = m \mid T_{ik} \geq m, \mathcal{H}_{ikm}) = &-2.5 + 0.5X_{1ik} + 0.4X_{2ik} \\ &+ 0.4L_{1ikm} + 0.3L_{2ikm} + 0.2L_{1ik,m-1} + 0.15L_{2ik,m-1} \\ &+ \beta_{\text{trt}}\mathbb{I}(S_{ik} \leq \tau, m \geq S_{ik}).\end{aligned} \tag{3}$$

Treatment affects the event hazard only if initiation occurs within the grace period. There is no additional direct site term in the event model. Site heterogeneity influences the marginal outcome distribution through patient mix and treatment initiation, not through an independent site outcome effect.

Within each interval, the simulation generates current L, then treatment initiation, and then the event. A patient may initiate and experience the event in the same interval. Initiation and covariate evolution stop after death.

5 Strategies, cloning, and artificial censoring

The two strategies are

$$\begin{aligned} g = 1: &\quad \text{initiate treatment within the grace period,} \\ g = 0: &\quad \text{do not initiate treatment within the grace period.} \end{aligned}$$

For the $g = 1$ clone,

$$C^1_{ik} = \begin{cases} M + 1, & S_{ik} \leq \tau, \\ \tau - 1, & S_{ik} > \tau. \end{cases}$$

For the $g = 0$ clone,

$$C_{ik}^0 = \begin{cases} S_{ik} - 1, & S_{ik} \le \tau, \\ M + 1, & S_{ik} > \tau. \end{cases}$$

Deviation occurs during the interval that triggers artificial censoring, so the clone is retained only through the preceding interval.

6 Scenario-specific oracle truth

6.1 Oracle target population

For each $(\tau, \beta_{\mathrm{trt}}, h)$ cell, the oracle simulates $N_{\mathrm{oracle}} = 3{,}000{,}000$ observations. Its site proportions match the finite-sample target population:

$$\pi_{kh} = \frac{n_{kh}}{\sum_{j=1}^{5} n_{jh}} = \frac{n_{kh}}{5000}. \tag{4}$$

Therefore, the oracle changes with the heterogeneity level not only because the site-specific DGP changes, but also because the target-population site mixture changes.

6.2 Oracle denominator and common numerator

The oracle uses the true initiation probability

$$q_{ikm}^{D,\mathrm{true}} = \mathrm{expit}(\alpha_{kh} + 0.8X_{1ik} - 0.3X_{2ik} + 0.5L_{1ikm} + 0.4L_{2ikm}).$$

Its common marginal numerator hazard is

$$q_m^{N,\mathrm{oracle}} = \frac{\sum_{k=1}^{5} \sum_i \mathbb{I}(S_{ik} = m, T_{ik} \ge m)}{\sum_{k=1}^{5} \sum_i \mathbb{I}(S_{ik} \ge m, T_{ik} \ge m)}. \tag{5}$$

This numerator automatically reflects unequal site sizes, different patient mixes, and different initiation practices.

6.3 Oracle weights and survival

Writing $q_m^N = q_m^{N,\mathrm{oracle}}$ and $q_{ikm}^D = q_{ikm}^{D,\mathrm{true}}$, the waiting and initiation factors are

$$r_{ikm}^{\mathrm{wait}} = \frac{1 - q_m^N}{1 - q_{ikm}^D}, \qquad r_{ikm}^{\mathrm{init}} = \frac{q_m^N}{q_{ikm}^D}.$$

The initiate-within-τ weight is

$$W_{ikm}^1 = \prod_{j=1}^{\min(m,\tau)} \left\{ \left(\frac{q_j^N}{q_{ikj}^D} \right)^{\mathbb{I}(S_{ik}=j)} \left(\frac{1 - q_j^N}{1 - q_{ikj}^D} \right)^{\mathbb{I}(S_{ik}>j)} \right\}, \tag{6}$$

while the do-not-initiate weight is

$$W_{ikm}^0 = \prod_{j=1}^{\min(m,\tau,S_{ik}-1)} \frac{1 - q_j^N}{1 - q_{ikj}^D}. \tag{7}$$

Factors are evaluated only while the patient remains eligible. After τ, the weights are carried forward. The current setting uses no effective weight truncation.

For strategy g, define

$$R^g_{ikm} = \mathbb{I}(T_{ik} > m-1, C^g_{ik} > m-1)$$

and

$$D^g_{ikm} = \mathbb{I}(R^g_{ikm} = 1, T_{ik} = m, T_{ik} \le C^g_{ik}).$$

The oracle hazard and survival curve are

$$\lambda^{g,\text{oracle}}_m = \frac{\sum_{k,i} W^g_{ikm} D^g_{ikm}}{\sum_{k,i} W^g_{ikm} R^g_{ikm}}, \qquad S^{g,\text{oracle}}(m) = \prod_{j=1}^{m}(1-\lambda^{g,\text{oracle}}_j).$$

At $M = 25$,

$$\psi_g = 1 - S^g(M), \qquad \text{RMST}_g = \sum_{m=0}^{M-1} S^g(m).$$

The reported contrasts are

$$\text{RD} = \psi_1 - \psi_0, \qquad \text{RR} = \frac{\psi_1}{\psi_0},$$

$$\text{OR} = \frac{\psi_1/(1-\psi_1)}{\psi_0/(1-\psi_0)}, \qquad \text{RMST}_{\text{diff}} = \text{RMST}_1 - \text{RMST}_0\,.$$

7 Federated CCW

7.1 Local denominator nuisance models

Each site constructs its own eligible person-interval dataset and fits

$$\begin{aligned}\text{logit}(\widehat{q}^{D,\text{local}}_{ikm}) = \widehat{\gamma}_{km} + \widehat{\beta}_{1k}X_{1ik} + \widehat{\beta}_{2k}X_{2ik} + \widehat{\beta}_{3k}L_{1ikm} + \widehat{\beta}_{4k}L_{2ikm} \\ + \widehat{\beta}_{5k}L_{1ik,m-1} + \widehat{\beta}_{6k}L_{2ik,m-1}.\end{aligned} \tag{8}$$

The interval effect $\widehat{\gamma}_{km}$ is represented by a factor for m. Every coefficient is estimated separately within site k. Thus, Fed-CCW does not borrow nuisance regression coefficients across sites. This model is more flexible than the DGP because it permits site-specific slopes and includes lagged L in the initiation model.

7.2 Common numerator and central estimation

Site k releases

$$N^{\text{init}}_{km} = \sum_i \mathbb{I}(S_{ik} = m, T_{ik} \ge m), \qquad N^{\text{eligible}}_{km} = \sum_i \mathbb{I}(S_{ik} \ge m, T_{ik} \ge m).$$

The center computes and broadcasts

$$\widehat{q}^N_m = \frac{\sum_k N^{\text{init}}_{km}}{\sum_k N^{\text{eligible}}_{km}}. \tag{9}$$

Each site then constructs the strategy weights using its local denominator and the common numerator. It releases weighted event and risk totals

$$DW_{km}^{g} = \sum_{i} W_{ikm}^{g} D_{ikm}^{g}, \qquad RW_{km}^{g} = \sum_{i} W_{ikm}^{g} R_{ikm}^{g}.$$

The central hazard is

$$\widehat{\lambda}_{m}^{g,\mathrm{fed}} = \frac{\sum_{k} DW_{km}^{g}}{\sum_{k} RW_{km}^{g}}, \tag{10}$$

followed by

$$\widehat{S}^{g,\mathrm{fed}}(m) = \prod_{j=1}^{m} (1 - \widehat{\lambda}_{j}^{g,\mathrm{fed}}).$$

This is one pooled weighted survival curve, not a meta-analysis of five local treatment-effect estimates.

For uncertainty estimation, global hazards and risk totals are returned to the sites. Sites calculate local individual influence contributions and release sums of squares. The implemented model-based variance treats the estimated weights as fixed and does not add nuisance-model estimation uncertainty.

8 Pooled CCW with fully site-stratified nuisance models

The aligned pooled estimator has all individual records centrally but fits Equation (8) separately within each site. It uses the same common numerator, artificial censoring, weight construction, weighted survival estimator, and influence-function calculation as Fed-CCW.

Its pooled weighted hazard is

$$\widehat{\lambda}_{m}^{g,\mathrm{strat}} = \frac{\sum_{k} \sum_{i} W_{ikm}^{g} D_{ikm}^{g}}{\sum_{k} \sum_{i} W_{ikm}^{g} R_{ikm}^{g}}.$$

Since the local sufficient statistics satisfy

$$DW_{km}^{g} = \sum_{i} W_{ikm}^{g} D_{ikm}^{g}, \qquad RW_{km}^{g} = \sum_{i} W_{ikm}^{g} R_{ikm}^{g},$$

we have

$$\widehat{\lambda}_{m}^{g,\mathrm{strat}} = \widehat{\lambda}_{m}^{g,\mathrm{fed}}.$$

Therefore, Fed-CCW and site-stratified pooled CCW have identical survival curves, treatment-effect estimates, and currently implemented model-based standard errors up to numerical precision. This comparison verifies lossless federation for the fully local nuisance-model specification.

9 Conventional pooled CCW with site fixed effects

The second pooled estimator fits one logistic initiation model to all pooled person-interval records:

$$\begin{aligned} \mathrm{logit}(\widehat{q}_{ikm}^{D,\mathrm{FE}}) = \widehat{\gamma}_{m} + \widehat{\delta}_{k} + \widehat{\beta}_{1} X_{1ik} + \widehat{\beta}_{2} X_{2ik} \\ + \widehat{\beta}_{3} L_{1ikm} + \widehat{\beta}_{4} L_{2ikm} + \widehat{\beta}_{5} L_{1ik,m-1} + \widehat{\beta}_{6} L_{2ik,m-1}. \end{aligned} \tag{11}$$

Here, $\widehat{\gamma}_m$ is a common interval effect, $\widehat{\delta}_k$ is a site fixed effect, and the $\widehat{\beta}$ vector is shared across sites. The model therefore permits residual differences in site initiation propensity while borrowing information across sites for the effects of patient history.

The common-slope restriction in Equation (11) is correct under the ArraySim5-4 DGP, because Equation (2) uses common covariate slopes and site-specific intercepts. The pooled fixed-effect model may therefore produce more stable nuisance estimates and weights, particularly at small sites. However, it is not the same statistical estimator as Fed-CCW's fully local nuisance-model approach.

After fitting Equation (11), this estimator uses the same common numerator and clone-censor-weight construction. Its hazard is

$$\widehat{\lambda}_m^{g,\mathrm{FE}} = \frac{\sum_{k,i} W_{ikm}^{g,\mathrm{FE}} D_{ikm}^{g}}{\sum_{k,i} W_{ikm}^{g,\mathrm{FE}} R_{ikm}^{g}}.$$

Because $W_{ikm}^{g,\mathrm{FE}}$ generally differs from the locally estimated Fed-CCW weight, this estimator need not equal Fed-CCW in finite samples. Any efficiency advantage reflects the additional common-slope assumption and cross-site information borrowing, not merely centralized data access.

10 Other comparison methods

The simulation also includes:

1. federated IPW without cloning;
2. federated unweighted per-protocol analysis; and
3. federated landmark IPW.

Together with the three CCW implementations, this gives six methods and eight reported estimands per method: ψ_1, ψ_0, RD, RR, OR, RMST_1, RMST_0, and $\mathrm{RMST}_{\mathrm{diff}}$.

11 Expected comparisons and positivity stress

The primary implementation identity is

$$\widehat{\theta}^{\mathrm{fed}} = \widehat{\theta}^{\mathrm{site\text{-}stratified\ pooled}}$$

for every replicate and CCW estimand θ, up to numerical precision. The conventional pooled fixed-effect estimator can differ because it estimates a common slope vector from all sites.

The high setting intentionally combines a small low-initiation site with a large high-initiation site and strongly separated covariate distributions. In the preflight sample with $\tau = 5$, observed initiation proportions ranged from approximately 0.018 to 0.756. Logistic fits may consequently warn about probabilities close to zero or one. These warnings represent the intended practical-positivity and local-information stress rather than a difference in the underlying causal estimand.

12 Monte Carlo evaluation

For each replicate r and estimand θ,

$$e_r(\theta) = \widehat{\theta}_r - \theta_{\text{oracle}}.$$

With $R = 100$ replicates,

$$\text{Bias}(\theta) = \frac{1}{R}\sum_{r=1}^{R} e_r(\theta), \qquad \text{RMSE}(\theta) = \left\{\frac{1}{R}\sum_{r=1}^{R} e_r(\theta)^2\right\}^{1/2}.$$

The aggregation also reports empirical standard deviation and confidence interval coverage when a model-based interval is available.